Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and spending patterns.

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Dataset at a Glance

3,900	18	$50.76	3.75
Purchases	Features	Avg Spend	Avg Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Category, Amount, Season, Size, Color

Behavior

Discounts, Frequency, Ratings, Shipping

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Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled 37 missing `Review Rating` values with category median

03

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Connected Python to PostgreSQL for SQL analysis

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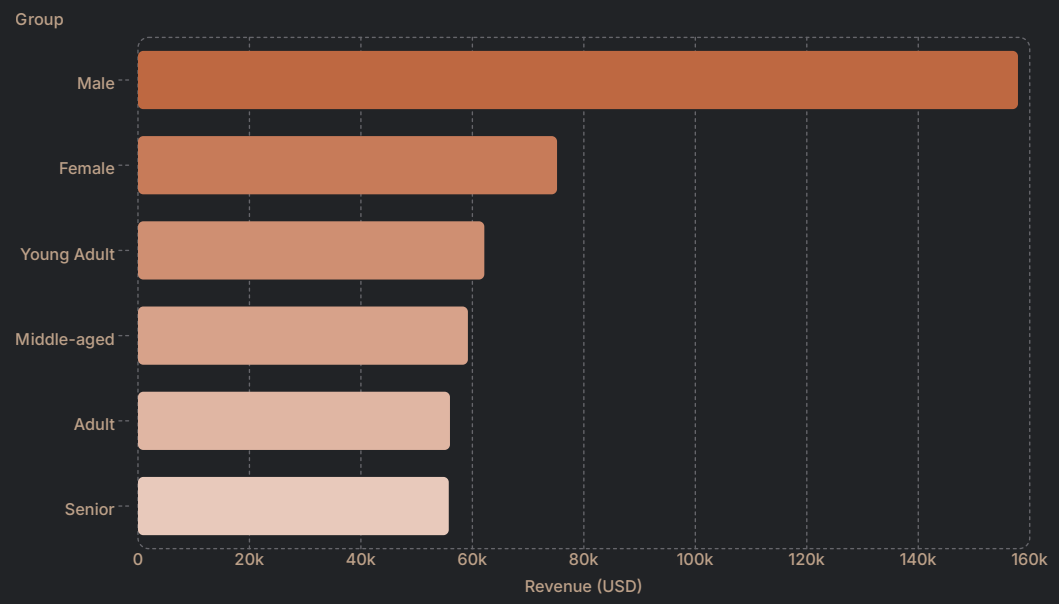
Revenue & Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults are the **top revenue age group**
- Express shipping users spend slightly more (**$60.48** vs. $58.46 Standard)
- 839 discount users still spent **above average**

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Top-Rated Products

Highest Avg Review Ratings

Product	Avg Rating
Gloves	3.86 ⭐
Sandals	3.84 ⭐
Boots	3.82 ⭐
Hat	3.80 ⭐
Skirt	3.78 ⭐

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Top Products by Category

👗 Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

💍 Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

👟 Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

🧥 Outerwear

- Jacket — 163
- Coat — 161

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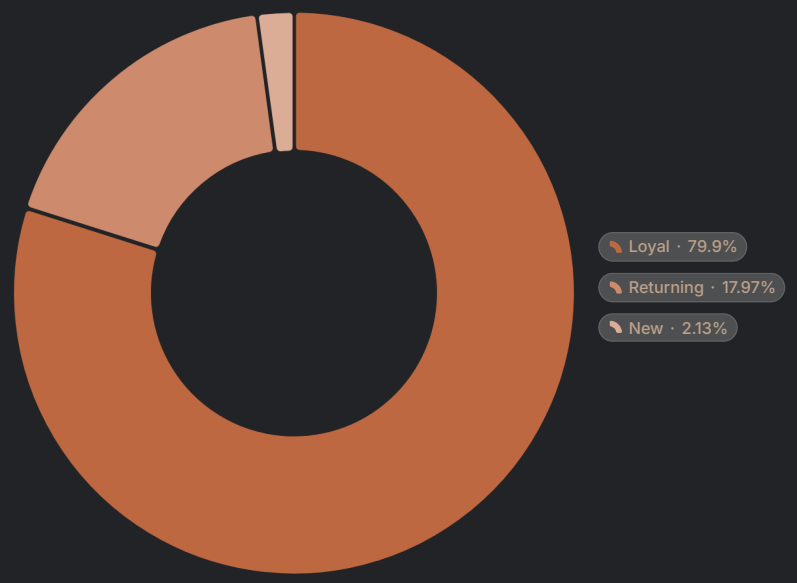
Customer Segmentation & Subscriptions



Subscription Status

Status	Customers	Avg Spend
Subscribers	1,053	$59.49
Non-subscribers	2,847	$59.87

Of repeat buyers (>5 purchases): **958 subscribe**, 2,518 do not.

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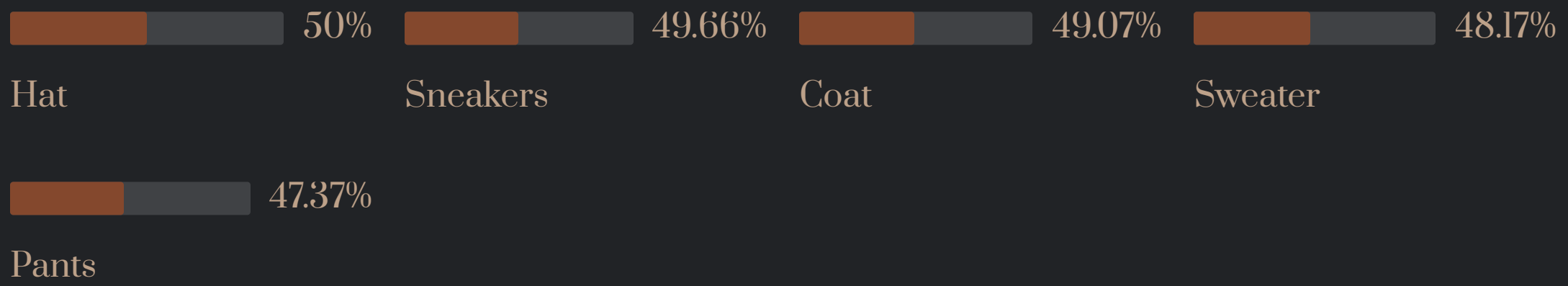
Discount-Dependent Products

Highest Discount Rates



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Power BI Dashboard

3.9K Customers

27% subscribers

$59.76 Avg Spend

Across all categories

3.75 Avg Rating

Filtered by gender, category & shipping

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Business Recommendations

→ **Boost Subscriptions**

Promote exclusive subscriber benefits

→ **Loyalty Programs**

Reward repeat buyers to grow the Loyal segment

→ **Review Discount Policy**

Balance sales boosts with margin control

→ **Targeted Marketing**

Focus on high-revenue age groups & express-shipping users

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